



**Department of Computer Science and Engineering**  
**Academic Year 2022 – 2023 (Even Semester)**

**Degree, Semester & Branch** : IV Semester B.E CSE  
**Course Code & Title** : CS3492 & Database Management Systems  
**Name of the Faculty member** : Mr. K.Vignesh Saravanan, AP / CSE

**Innovative Practice Description**

- **Unit / Topic:** Unit III / Transactions / Deadlock Handling
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Flipped Classroom
- **Justification:**

Flipping the classroom is an inverting the classroom approach to teaching. In this approach, the traditional in-class teaching is “flipped” to better meet the needs of individual learners. Students gain control of the learning process through studying course material outside of class, using readings, pre-recorded video lectures. It helps the faculty/lecturer to redefine in-class activities and include homework problems and keep the students engaged in the content.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  - **Plan:** Basics of Deadlock are already learnt in the subject operating systems and few topics related to DBMS is given as self-learning through self-exploration.
  - **Identify and Share:** Related materials are identified and I posted relevant references/materials/notes of deadlock mechanisms to the students and given as self-learning. The Figure.1 shows the activities / discussion as a team on the self-exploration topics.
  - **Evaluate:**
    - I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/present in class. The Figure.1 images shows the student Subana Devi and Haridharsana explaining the self-explored concepts on deadlocks.
    - Make the students to find out the answer by their own by learning.

• **CO – PO / PSO mapping:**

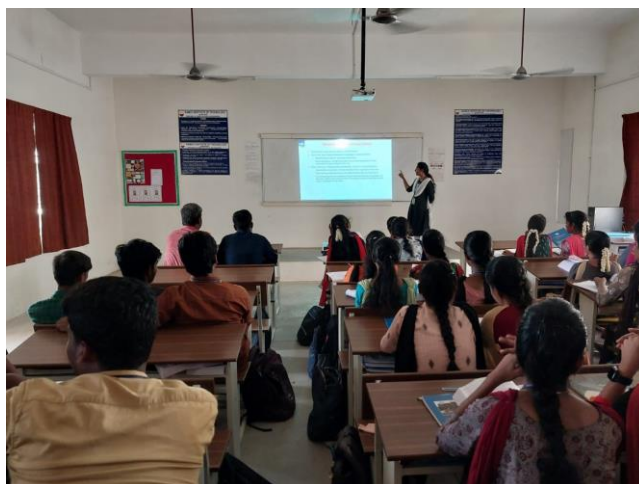
| CO  | PO1 | PO2 | PO3 | PO7 | PO8 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 3   | 3   | 1   | 1   | 1   | 1    | 2    | 3    |

(1 – Low      2 – Moderate      3 – High)

• **PO / PSO mapped:**

| Innovative practice                  | PO1                                                                  | PO2                                                    | PO3                                                                              | PO9                                                                                        | PO10                                                                               | PO12                                                             |
|--------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------|
|                                      | 2                                                                    | 2                                                      | 3                                                                                | 3                                                                                          | 3                                                                                  | 2                                                                |
| <b>Justification for correlation</b> | Apply basic Knowledge and fundamentals in Transactions and deadlocks | Identify the need of deadlock prevention and detection | Able to design and develop the mechanism for the deadlock prevention & avoidance | Functional individually in identifying the protocols for deadlock prevention and avoidance | Communicate / share the ideas with other students in written and oral presentation | Ability to reproduce the contents gathered through self-learning |

• **Images / Screenshot of the practice:**



**Figure.1 – Screenshots of the students activities in the Flipped classroom innovative practice**

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Student felt good, since they can study at their own pace/time.
    - They felt that through such learning, they can explore more.

- ❖ ***Benefit of the practice:***

- More one-to-one time between teacher and student.
    - More collaboration time for students.
    - Students learn at their own pace.
    - Practical things – like missing class due to illness – become less problematic.
    - It encourages students to come to class prepared.

- ❖ ***Challenges faced in implementation:***

- Relies on student preparation – few students did not come prepared.
    - The depth of the subject can be dictated by the student themselves or the group the student is working with.
    - The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

**References:**

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- ❖ [https://en.wikipedia.org/wiki/Flipped\\_classroom](https://en.wikipedia.org/wiki/Flipped_classroom)
- ❖ [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>